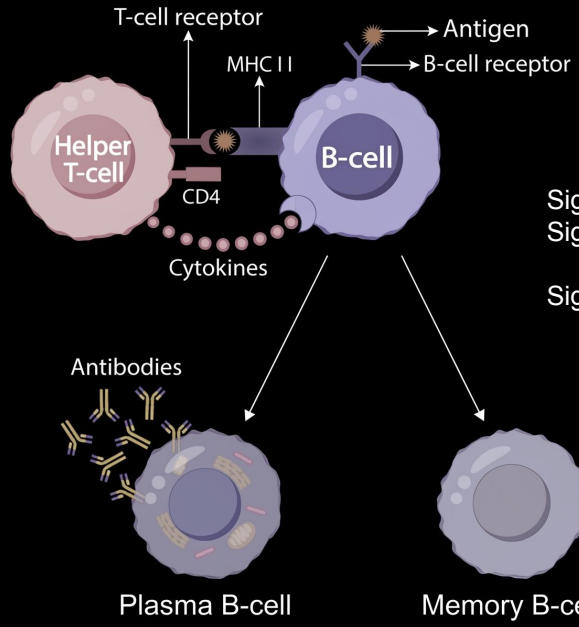
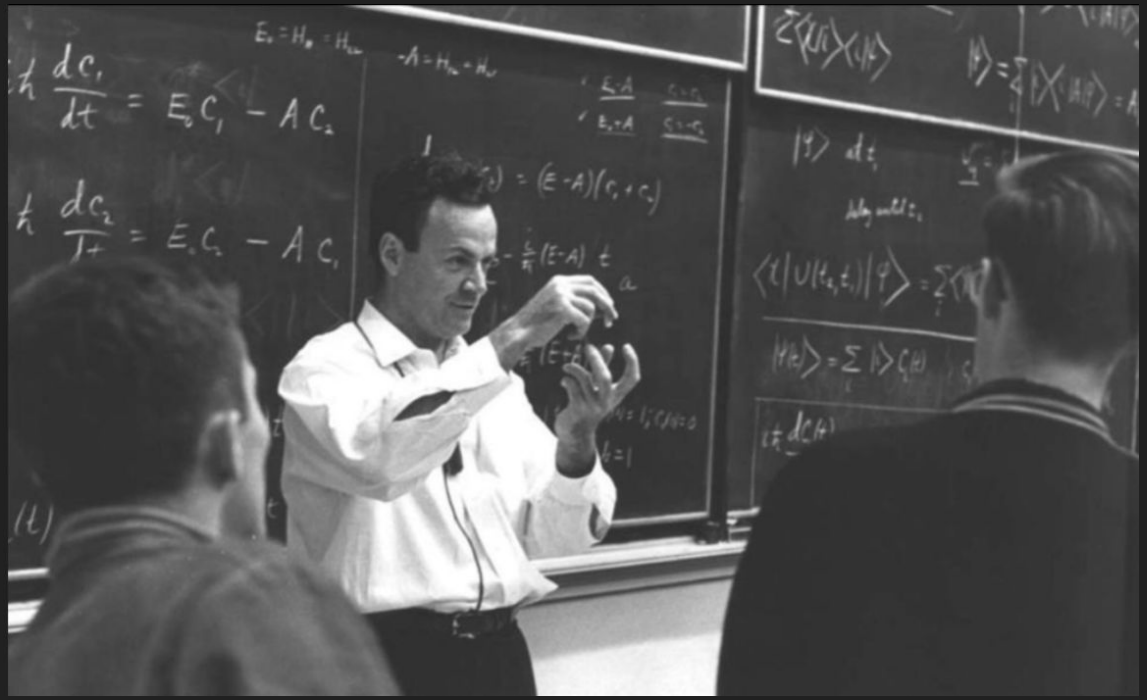
Immune System - High Level Process



Signal 1 - Antigen Recognition
Signal 2 - Co-Stimulation
- Somatic HyperMutation

Signal 3 - Cytokine Signaling and Clonal Selection

Richard Feynman



“The statements of science are not of what is true and what is not true, but statements of what is known with different degrees of certainty”

Emergent System Symmetries *(from Cosmos to Cells)*

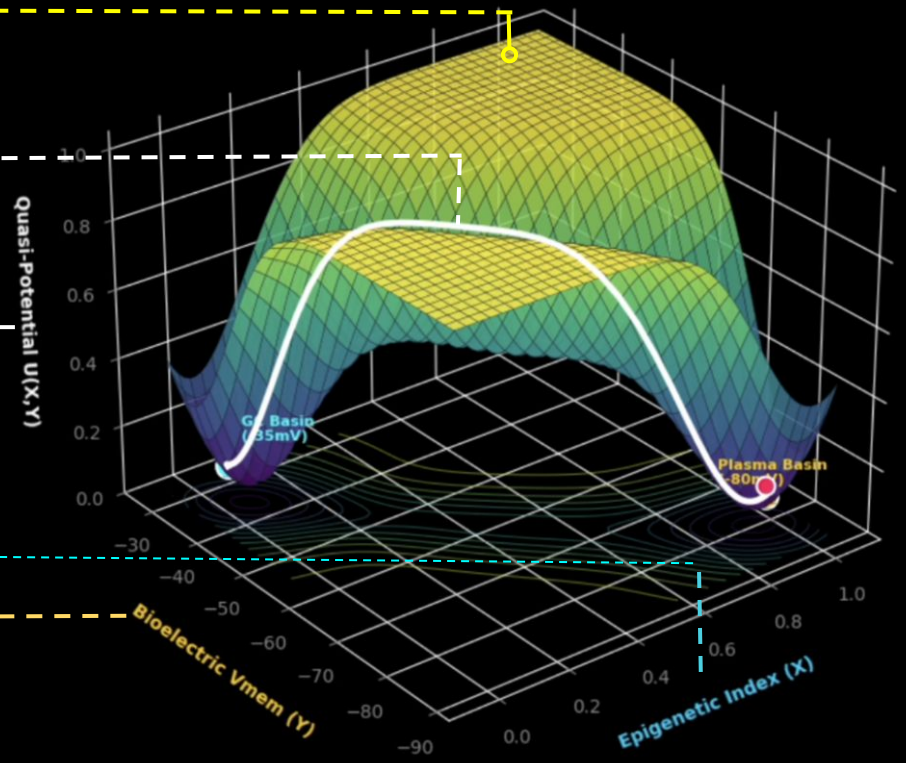
	Field Equation (Landscape)	The Equation of Motion (Path)
Cosmic Level	$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$	$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0$
Biological Level	$U(\mathbf{y}) = -\ln P_{\text{ss}}(\mathbf{y})$	$\frac{d\mathbf{y}}{dt} = -\mathbf{D}\nabla U(\mathbf{y}) + \mathbf{J}_{\text{curl}}(\mathbf{y}) + \boldsymbol{\xi}(t)$

Biological Level 1: Landscape and Path

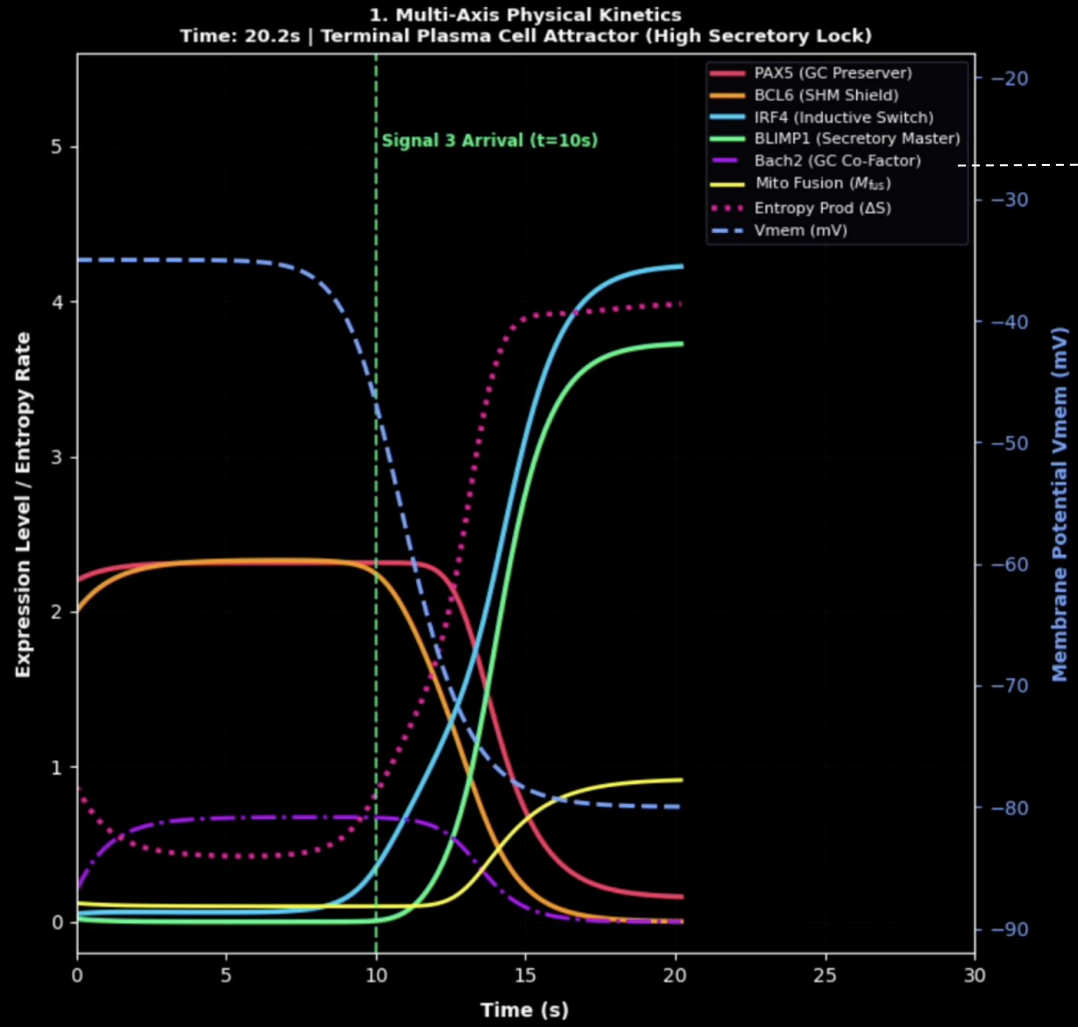
$$U(\mathbf{y}) = -\ln P_{\text{ss}}(\mathbf{y})$$

$$\frac{d\mathbf{y}}{dt} = -\mathbf{D}\nabla U(\mathbf{y}) + \mathbf{J}_{\text{curl}}(\mathbf{y}) + \boldsymbol{\xi}(t)$$

$$\mathbf{y} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix} = \begin{bmatrix} [\text{PAX5}] \\ [\text{BCL6}] \\ [\text{IRF4}] \\ [\text{BLIMP1}] \\ [\text{BACH2}] \\ V_{\text{mem}} \\ M_{\text{fus}} \end{bmatrix}$$



Level 2: Cell Kinetics



$$\mathbf{y} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix} = \begin{bmatrix} [\text{PAX5}] \\ [\text{BCL6}] \\ [\text{IRF4}] \\ [\text{BLIMP1}] \\ [\text{BACH2}] \\ V_{mem} \\ M_{fus} \end{bmatrix}$$

$\triangle$ S Entropy Production

Level 2: Cell Kinetics (Equations)

$$\mathbf{y} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix} = \begin{bmatrix} [\text{PAX5}] \\ [\text{BCL6}] \\ [\text{IRF4}] \\ [\text{BLIMP1}] \\ [\text{BACH2}] \\ V_{mem} \\ M_{fus} \end{bmatrix}$$

$$\begin{bmatrix}
y_0 = \dfrac{d[\text{PAX5}]}{dt} = \dfrac{2.2}{1 + \text{BLIMP1}^2} - 0.8\,[\text{PAX5}] \\
y_1 = \dfrac{d[\text{BCL6}]}{dt} = 2.5 \left(\dfrac{\text{PAX5}^2}{1.0 + \text{PAX5}^2}\right)\left(\dfrac{1}{1 + \text{BLIMP1}^2}\right)\left(\dfrac{1}{1 + \text{IRF4}^2}\right) - 0.9\,[\text{BCL6}] \\
y_2 = \dfrac{d(\text{IRF4})}{dt} = \left(1.6\,S_3 + \Phi_{\text{NFAT}}\right) \cdot R_{\text{shield}} + 0.6 \left(\dfrac{\text{IRF4}^2}{1.0^2 + \text{IRF4}^2}\right) - 1.05 \cdot \text{IRF4} \\
y_3 = \dfrac{d[\text{BLIMP1}]}{dt} = 3.8 \left(\dfrac{\text{IRF4}^3}{1.0^3 + \text{IRF4}^3}\right)\left(\dfrac{1}{1 + \text{BCL6}^2}\right)\left(\dfrac{1}{1 + \text{BACH2}^2}\right) - 1.0\,[\text{BLIMP1}] \\
y_4 = \dfrac{d[\text{BACH2}]}{dt} = 0.8 \left(\dfrac{\text{BCL6}^2}{1.0 + \text{BCL6}^2}\right)\left(\dfrac{1}{1 + \text{BLIMP1}^2}\right) - 1.0\,[\text{BACH2}]
\end{bmatrix}$$

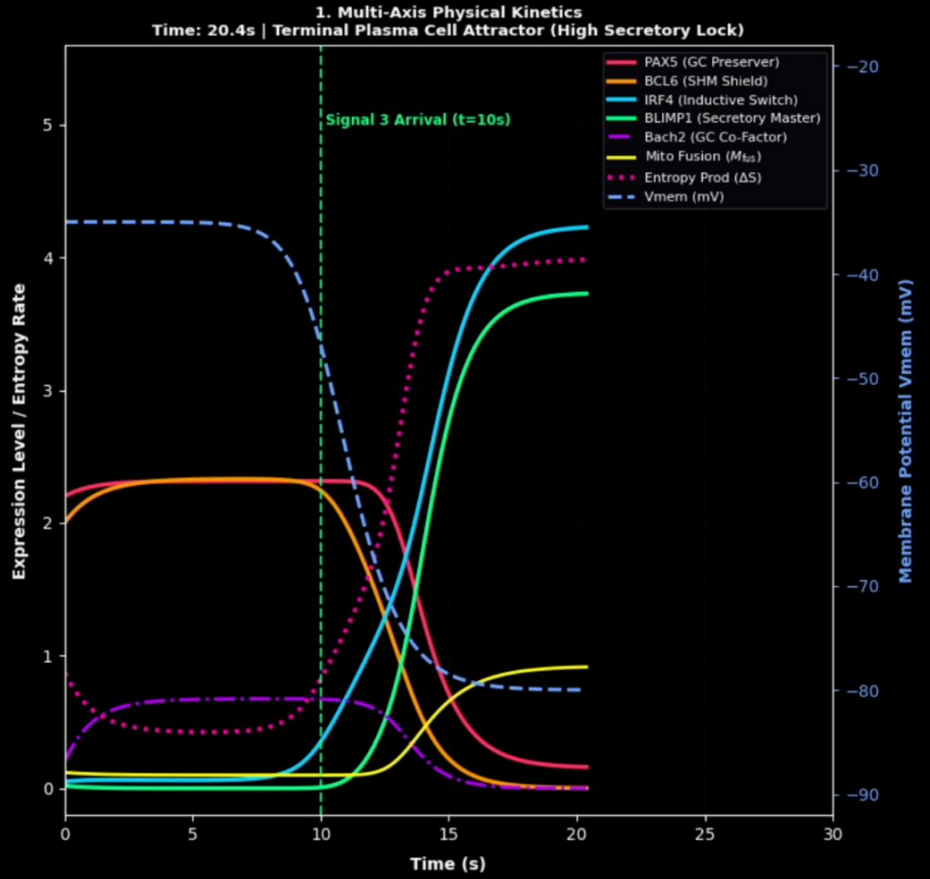
$$\begin{bmatrix} y_5 = \dfrac{dV_{\text{mem}}}{dt} = 0.85 \cdot \left((-35.0 - 45.0\,S_3) - V_{\text{mem}}\right) \end{bmatrix}$$

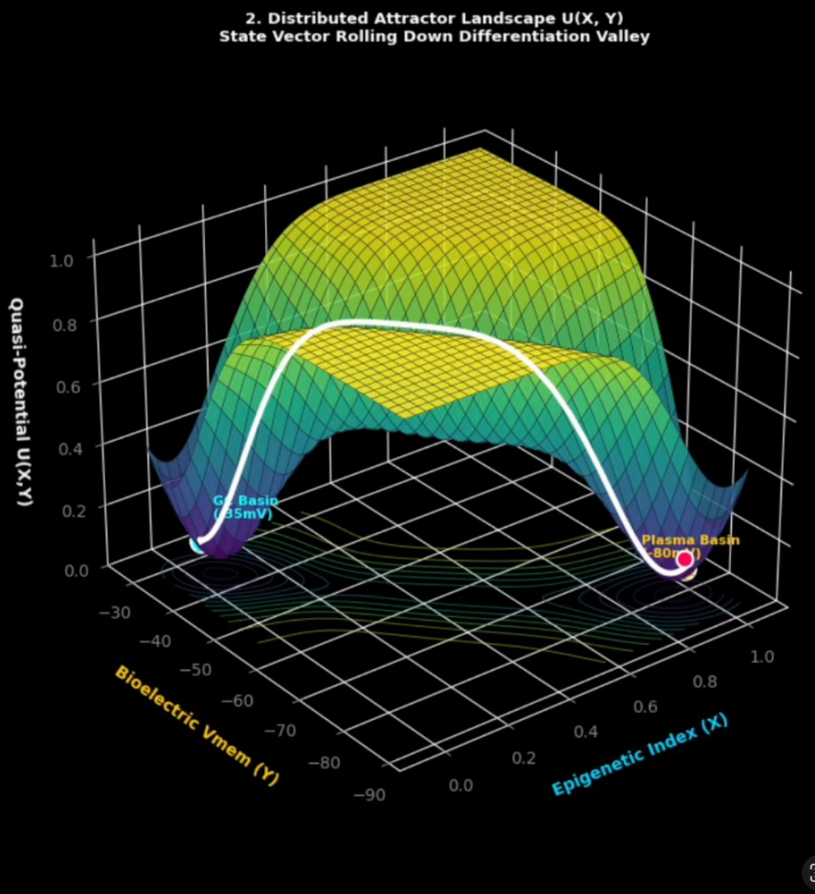
$$\begin{bmatrix} y_6 = \dfrac{dM_{\text{fus}}}{dt} = 0.75 \cdot \left(\left[0.10 + 0.88 \left(\dfrac{\text{BLIMP1}^2}{1.0^2 + \text{BLIMP1}^2 + 10^{-6}}\right)\right] - M_{\text{fus}}\right) \end{bmatrix}$$

$$\Delta S\,(t) \;=\; 0.45 \sum_{t=0}^{4} |\dot{y}| \;+\; \left[0.30 \quad + 0.70 \max\left(0, \tfrac{(V_{mem} - (-35))}{-45}\right)\right] M_{fus} \;+$$

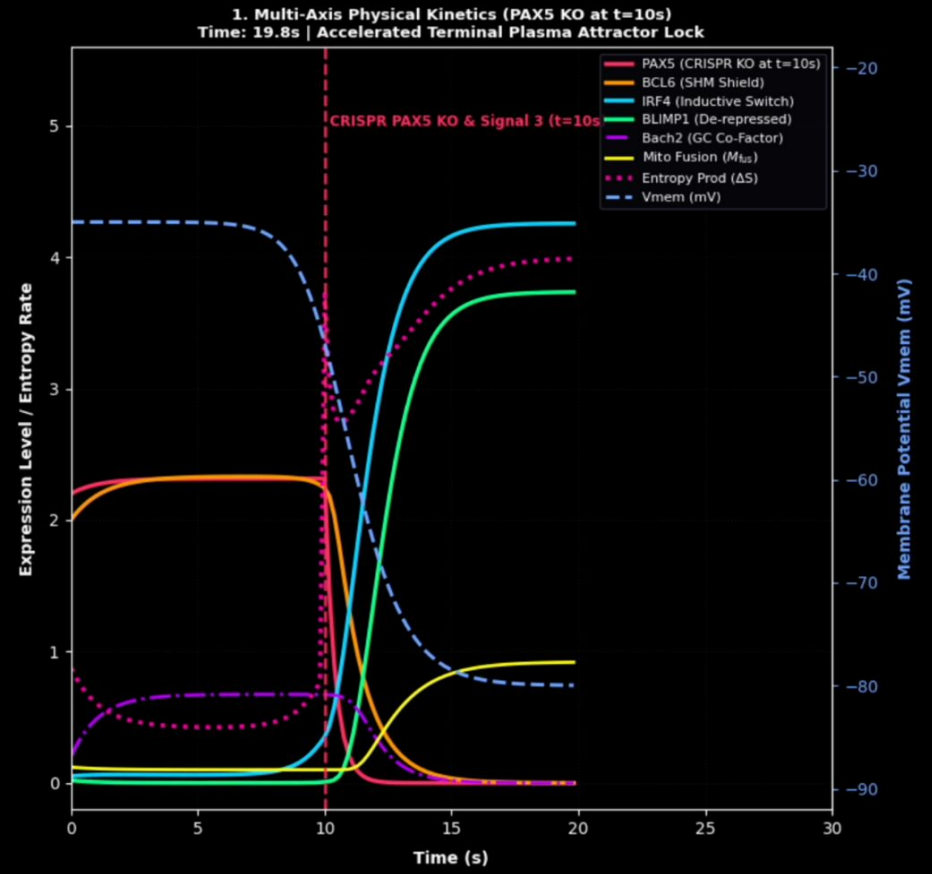
$$\left[0.8\,(M_{fus})^2 \;+\; 1.2\,M_{fus}\right] \;+\; 0.55\,BLIMP1$$

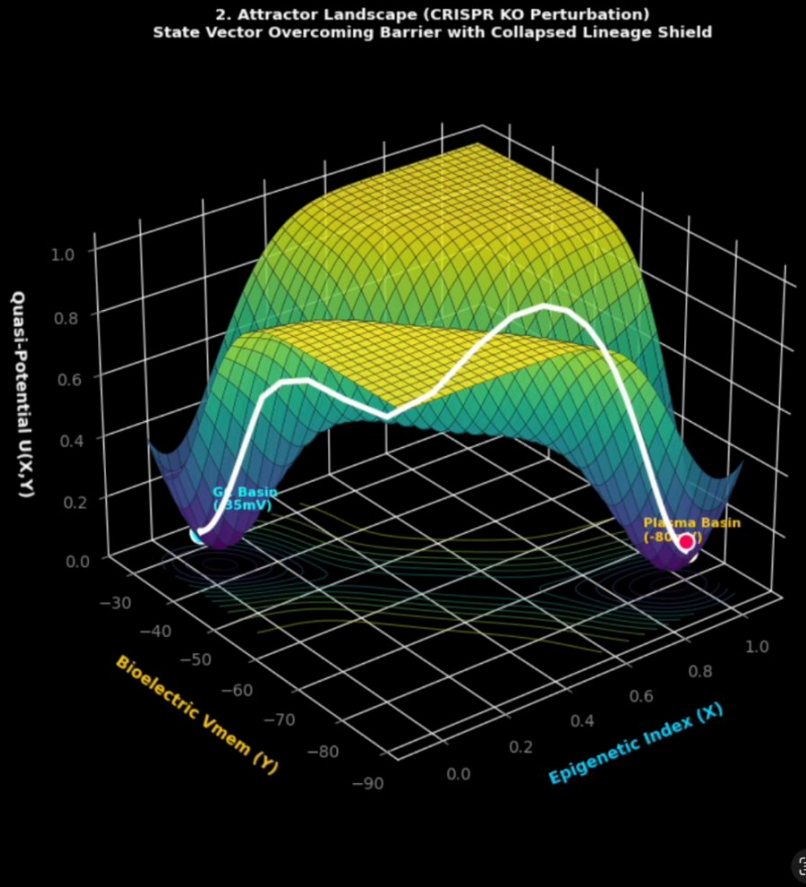
Movie: Normal Immune System Process





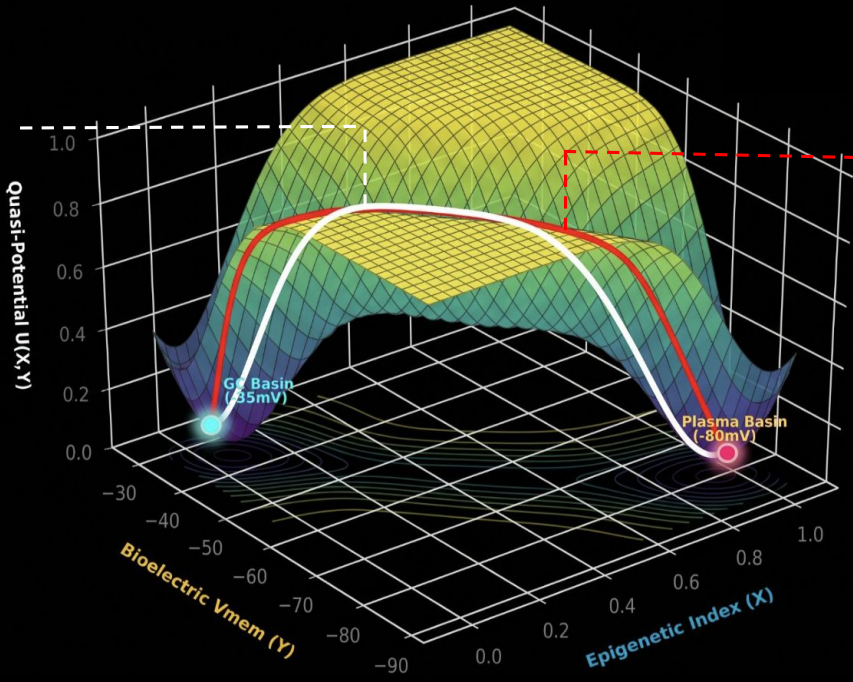
Movie: Knockout Immune System Process





Analysis Approach Comparison

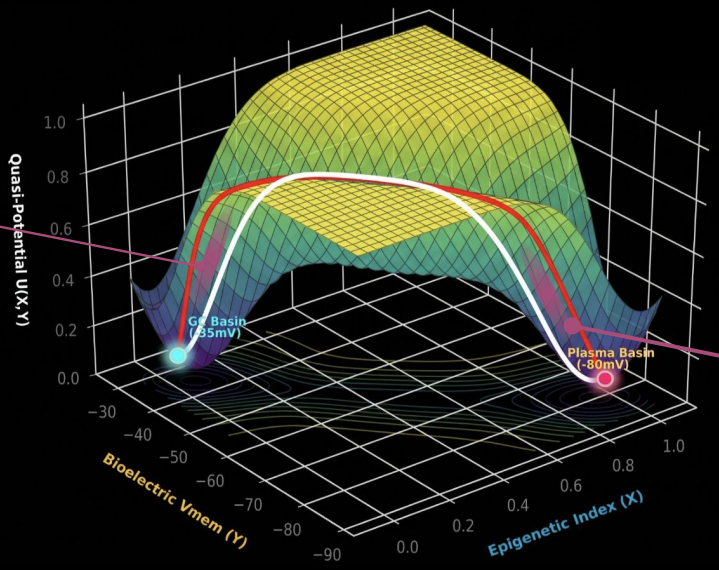
Emergent Model



Classic GRN Models

Opportunity in the Gaps

- Cell Exhaustion
- Synthetic ceilings
- ER stress
- Premature Apoptosis
- Expression Failures



- Unintended fate drift *(off-target errors)*
- Loss of potency
- Functional decline and loss of persistence, physiological performance lag *(esp. within harsh tissue environments - e.g., tumors)*